

Embodied AI and Cognitive Psychometrics: Evaluating a Simulated *Drosophila* Connectome in a Web-Based Language Acquisition Environment

The intersection of embodied artificial intelligence, whole-brain connectomics, and advanced psychometric evaluation represents a critical frontier in understanding the sensorimotor and cognitive basis of learning. The programmatic exploration of these domains—often conceptualized within initiatives focused on psychometric and neural learning interfaces—aims to bridge the gap between biological neuroanatomy and machine learning performance metrics. The recent development of anatomically detailed, physics-based simulations of the fruit fly (*Drosophila melanogaster*), coupled with complete synaptic-resolution brain mapping, provides an unprecedented substrate for *in silico* behavioral experiments. This report details a comprehensive theoretical and methodological framework for evaluating a simulated *Drosophila* as it interacts with a web-based language acquisition platform, conceptually modeled after the Duolingo smartphone application, rendered entirely within a 3D browser environment.

To measure, guide, and assess the simulated organism's performance in pattern recognition, sequential logic, and proto-reading tasks, a highly sophisticated ensemble of algorithms is required. This framework integrates high-fidelity biomechanical rendering, connectomic graph modeling (FlyGM), advanced sequence modeling (xLSTM and Decision Transformers), and rigorous psychometric and memory-tracking algorithms (G-DINA, Half-Life Regression, Dynamic Bayesian Networks, and the Many-Facet Rasch Model). Furthermore, the infrastructure requires specialized telemetry (xAPI) and network architectures (Science DMZ) to handle the immense data throughput generated by the physics engine and the neural controller. This document synthesizes these disparate algorithms into a unified evaluative framework, establishing how they collectively measure the learning trajectory of a virtual biological agent interacting with a digital interface.

The Biomechanical and Neural Substrate of the Virtual *Drosophila*

The foundation of this experimental paradigm is the virtual *Drosophila* body model, a highly accurate biomechanical simulation developed through collaborative efforts between Google DeepMind and the HHMI Janelia Research Campus¹. To assess a simulated fly's ability to learn, the agent must possess a physical embodiment that enforces the same kinematic and sensory constraints experienced by biological organisms.

The MuJoCo flybody Physics Simulation

The flybody repository provides a whole-body model of the female *Drosophila melanogaster* rendered as an Extensible Markup Language (XML) model for the MuJoCo physics simulator¹.

Unlike stylized, generic multi-legged robots that approximate movement, this model is constructed from actual exoskeleton geometry, joint arrangements, and muscle-actuator dynamics down to individual leg segments².

The model requires a 59-dimensional action space simply to coordinate walking, representing the independent degrees of freedom across the fly's legs and body¹. To interact with a simulated smartphone screen displaying language tasks, the fly must execute highly coordinated terrestrial locomotion and vision-guided navigation². The MuJoCo physics engine calculates fluid dynamics for wing forces and adhesion actuators for the tarsal segments, allowing the fly to physically grip surfaces, including the simulated glass of a digital screen⁴. This level of embodied realism ensures that any cognitive learning measured by the psychometric algorithms is grounded in actual physical interaction, preventing the neural controller from exploiting unrealistic physics loopholes.

Connectome-Constrained Control: The FlyGM Architecture

Controlling a 59-degree-of-freedom body to achieve goal-directed behavior—such as selecting a specific phonetic symbol on a screen—requires a sophisticated neural policy. Conventional artificial intelligence applications rely heavily on standard Multi-Layer Perceptrons (MLPs), which lack biological plausibility. Instead, this paradigm utilizes the Fly-connectomic Graph Model (FlyGM)⁷. The FlyGM instantiates the complete adult *Drosophila* whole-brain connectome, a massive dataset mapped by the FlyWire consortium detailing over 130,000 neurons and approximately 50 million synapses⁸.

In the FlyGM architecture, neurons are strictly partitioned based on biological information flow, transforming the static wiring diagram into a directed message-passing graph⁷. Table 1 delineates the functional categorization of the neural nodes within the FlyGM network, illustrating how the simulated fly processes interface stimuli and translates them into biomechanical interactions.

Neuron Classification	Anatomical Role in FlyGM	Function in the Language Acquisition Paradigm
Afferent Neurons	Visual, olfactory, and mechanosensory receivers.	Process pixels from the rendered smartphone screen and proprioceptive feedback from tarsal screen contact.
Intrinsic Neurons	Central brain interneurons, mushroom bodies, and central complex.	Mediate recurrent computations, associative memory formation, and action selection (choosing which symbol to approach).

Efferent Neurons	Descending motor neurons connecting to the ventral nerve cord.	Generate the high-frequency torque commands that drive the 59 actuators in the MuJoCo body model.
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By utilizing the connectome as a structural inductive bias, the FlyGM demonstrates superior sample efficiency and lower kinematic error compared to randomly wired or non-biological networks⁷.

Mitigating the "Digital Sphinx" Conundrum

The convergence of whole-body physics simulations and connectome-based neural control raises profound questions regarding the interpretability of *in silico* behavior. A central methodological challenge in this space is what computational neuroscientists term the "Digital Sphinx" problem¹².

Recent attempts to embed the neural connectome of a nematode (*C. elegans*) into the biomechanical body model of a fruit fly demonstrated that Deep Reinforcement Learning (DRL) algorithms can coerce fundamentally incompatible neural architectures into generating realistic walking kinematics¹². However, because the *C. elegans* connectome evolved for undulatory movement through soil rather than articulated hexapod locomotion, the resulting model was biologically meaningless. The network merely acted as an arbitrary recurrent neural network (RNN) with sufficient dynamics to overfit the task, meaning behavioral fidelity was achieved entirely without biological fidelity¹².

This framework explicitly mitigates the Digital Sphinx paradox by ensuring structural and evolutionary congruence. By utilizing the FlyGM, the architecture strictly preserves the *Drosophila* sensorimotor routing—from the ommatidia afferents to the leg and wing efferents—guaranteeing species-appropriate structural inductive biases⁸. Consequently, when psychometric algorithms evaluate the fly's learning trajectory, researchers can be confident that the observed cognitive adaptations are grounded in a valid biological topology rather than an overfitted algorithmic artifact.

Sensory Constraints and Reinforcement Mechanisms

Evaluating the fly's ability to learn proto-reading tasks requires a deep understanding of its sensory limitations and the biological mechanisms that drive reinforcement. The simulated environment must be explicitly tailored to the optical realities of the *Drosophila* compound eye and its dopaminergic reward pathways.

Visual Acuity and Microsaccadic Sampling

A critical physical challenge in teaching a virtual fly to "read" is the inherent limitation of its visual system. The *Drosophila* compound eye is tiled by approximately 750 ommatidia arranged in a hexagonal grid, providing a nearly panoramic field of view but severely limited static spatial resolution¹³. Each ommatidium acts as a miniature lens with an aperture of roughly 16

micrometers and a focal length of 20 micrometers, collecting light from a visual angle of approximately 5° (the inter-ommatidial angle)¹⁴. Based on classical optical theory, this static resolution is mathematically insufficient to resolve the fine, high-frequency spatial details required to differentiate complex graphemes on a standard digital interface¹⁴. However, biological flies achieve hyperacute vision—the ability to resolve spatial details significantly finer than their optical sampling limit—through active image motion and microsaccadic behaviors¹³. Wide-field motion-sensitive neurons in the visual system, known as Lobula Plate Tangential Cells (LPTCs), integrate local motion signals to produce highly sensitive optic flow responses¹⁶.

To perceive the letters on the simulated Duolingo interface, the virtual fly cannot passively stare at the screen. The physics simulation must require the fly to actively move its body and head across the text. This allows the FlyGM visual afferents to process rapid contrast changes over time, mimicking biological microsaccadic sampling and transforming temporal resolution into spatial acuity¹³. Furthermore, the rendering of the smartphone app within the simulation must account for the geometric convergence of the ommatidial optical axes, which creates distance-dependent differences in receptive field overlap, requiring the fly to maintain an optimal viewing distance of 1 to 10 millimeters from the simulated screen to accurately resolve depth and shape¹⁹.

The Dopaminergic PAM Cluster Reward Architecture

In the physical Duolingo application, human learners receive gamified rewards, such as auditory chimes and visual animations, to reinforce correct answers. For a simulated *Drosophila*, these human-centric rewards are meaningless and must be translated into biologically valid reinforcement signals.

In the biological fruit fly, the acquisition of associative memories—specifically appetitive memories linking visual or olfactory cues to a reward—is primarily mediated by dopaminergic neurons located in the protocerebral anterior medial (PAM) cluster²⁰. These neurons signal the rewarding properties of sugar ingestion²⁰. Research indicates that the PAM cluster is functionally heterogeneous; distinct subsets of these dopamine neurons are responsible for signaling rewards that induce short-term memory (STM), which decays within hours, versus long-term memory (LTM), which requires repeated consolidation and remains stable over extended periods²¹.

In the simulation framework, the human gamification mechanics are replaced by targeted neurochemical simulations. When the fly correctly navigates to the matching symbol or word on the smartphone screen, the environment programmatically injects a simulated dopamine transient directly into the corresponding PAM cluster nodes within the FlyGM connectome²¹. By differentially activating the STM-associated and LTM-associated PAM subsets based on the spacing and repetition of the tasks, the simulation bridges the gap between digital software mechanics and connectome-level reinforcement learning, providing a foundation for evaluating memory retention over time.

Infrastructure: Web-Based Simulation and

High-Frequency Telemetry

Deploying this highly complex, 3D physics-based assessment requires a robust infrastructure capable of rendering the environment and capturing granular, high-frequency behavioral data without inducing computational bottlenecks.

Browser-Based MuJoCo via WebAssembly

The visual and physical environment is constructed using advanced WebGL and WebGPU technologies, operating directly within a standard web browser²³. The MuJoCo physics engine, conventionally executed natively on desktop environments, is compiled to WebAssembly (WASM). This allows the entire 59-DoF physics simulation, along with the neural network inference required to drive the FlyGM, to execute directly in the client browser with minimal latency²³.

To achieve the necessary performance, cross-origin isolation utilizing Cross-Origin Opener Policy (COOP) and Cross-Origin Embedder Policy (COEP) headers is implemented. This enables SharedArrayBuffer threading, which drastically reduces forward-pass computational times²³. By leveraging multithreading, the simulated fly can react to the dynamically changing smartphone screen in real-time, relying on genuine simulation physics and neural policy responses rather than pre-baked kinematic animations²³.

Telemetry Streaming: xAPI and the Science DMZ

Evaluating the fly's learning trajectory requires tracking more than just discrete correct or incorrect answers. The psychometric algorithms demand continuous joint kinematics, visual sampling paths, and internal neural state activations at millisecond resolution.

To standardize this data capture, the system utilizes the Experience API (xAPI, formally standardized as IEEE 9274.1.1)²⁵. Traditional xAPI statements take a semantic structure, such as [Actor] [Verb] [Object]. In this high-frequency context, xAPI profiles are extended to encapsulate simulation states, recording the continuous Cartesian coordinates of the fly's center of mass, the activation states of its tarsal adhesion actuators, and specific ommatidial firing patterns at each simulation step²⁸.

Because a physics simulation running at 800 frames per second generates massive volumes of telemetry, routing this data through conventional enterprise firewalls would induce unacceptable latency and packet loss³⁰. Therefore, the data pipeline is architected around a Science DMZ. A Science DMZ is a specialized network sub-architecture explicitly designed for high-volume scientific data transfers, optimizing for a moderate number of high-speed flows rather than general-purpose business traffic³⁰. By bypassing stateful firewall inspections using dedicated high-performance Data Transfer Nodes (DTNs), the xAPI telemetry streams directly and securely into a remote Learning Record Store (LRS) for real-time psychometric processing²⁹.

Sequence Modeling for Adaptive Curriculum

Navigation

While the static connectome (FlyGM) provides the biological routing for sensory-motor transformations, driving complex, long-horizon tasks such as interacting with a sequentially

structured language curriculum requires machine learning models capable of adaptive planning and robust memory retention. To achieve this, the evaluative framework integrates Extended Long Short-Term Memory (xLSTM) networks and Decision Transformers to assist and interpret the biological controller.

The xLSTM Architecture for High-Capacity Visual Association

Classical Long Short-Term Memory (LSTM) networks suffer from inherently limited storage capacity because their hidden cell states are constrained to scalar or vector representations. This architectural bottleneck forces high-dimensional sensory information to be aggressively compressed into single values, leading to rapid information loss³¹. This limitation is fatal when evaluating an agent tasked with memorizing the complex mapping of distinct visual shapes (graphemes) to specific semantic rewards over long operational contexts³¹.

The xLSTM architecture, introduced by Beck et al. (2024), resolves this bottleneck by substituting the scalar memory with a sophisticated matrix memory structure, known specifically

as the mLSTM³². The mLSTM cell replaces the standard scalar cell state c_t with a matrix $C_t \in \mathbb{R}^{d \times d}$ ³⁵. It operates using a covariance update rule, allowing the network to store high-capacity key-value associations critical for pattern recognition³³. The mathematical formulation governing the mLSTM cell state update is defined as:

$$C_t = f_t C_{t-1} + i_t v_t k_t^T$$

where f_t represents the forget gate, i_t represents the input gate, v_t denotes the value vector, and k_t denotes the key vector³¹.

Furthermore, xLSTM abandons traditional sigmoid activations in favor of exponential gating mechanisms. To prevent numerical instability and overflow during the exponential calculations,

the architecture introduces dedicated normalizer states (n_t) and stabilizer states (m_t), ensuring proper scaling while maintaining mathematical equivalence³¹.

To execute the xLSTM inference efficiently within the physics simulation loop, Tiled Flash Linear Attention (TFLA) is employed. TFLA is a hardware-aware kernel algorithm that parallelizes the linear RNN computations over chunks of the input sequence³⁹. By keeping intermediate states in high-speed SRAM, TFLA maximizes WebGPU memory bandwidth utilization, providing linear compute scaling with respect to the sequence length³⁹. In the context of the simulated fly, the xLSTM acts as a high-fidelity associative memory buffer, mapping the sequential visual streams generated by the fly's saccadic inputs directly to the historical dopamine rewards, allowing researchers to evaluate the agent's associative recall capabilities.

Adaptive Sequencing via Decision Transformers

To train and evaluate the fly as it navigates the language curriculum, the learning process is framed not as standard Temporal Difference (TD) reinforcement learning, but as a sequence modeling problem via a Decision Transformer (DT)⁴².

The Decision Transformer operates autoregressively on trajectories composed of states, actions, and rewards, conditioned heavily on a parameter known as the Return-to-Go (RTG)⁴⁵.

In the Duolingo simulation environment, the state $\mathbf{s}_t$ comprises the visual input from the ommatidia and the proprioceptive joint kinematics; the action $\mathbf{a}_t$ is the 59-dimensional motor command sent to the body actuators; and the RTG $\hat{R}_t$ is the target cumulative PAM cluster dopamine reward⁴³. The DT predicts the optimal motor action necessary to achieve the desired future reward⁴³.

To optimize computational efficiency within the web environment, a Decoupled Decision Transformer (DDT) variant is utilized. The DDT streamlines the architecture by recognizing a critical redundancy in standard DTs: feeding the entire historical sequence of RTGs into the attention mechanism is theoretically unnecessary, as only the most recent RTG fundamentally affects the next action prediction⁴⁵. The DDT processes only the observation and action sequences through the Transformer layers, relying on the latest RTG to guide action generation⁴⁵. By dynamically modulating the RTG input based on the fly's prior success rates, the DDT acts as an adaptive curriculum sequencer, continuously adjusting the complexity of the reading tasks displayed on the smartphone screen to perfectly match the fly's current inferred cognitive capability.

Cognitive Diagnosis and Psychometric Evaluation

As the virtual fly physically interacts with the digital language platform, assessing its internal "learning" requires quantitative psychometric models. A core innovation of this framework is the adaptation of human educational measurement tools—specifically Cognitive Diagnosis Models (CDMs)—to evaluate the cognitive state of non-human embodied agents.

Latent Attribute Mastery: The G-DINA Model

To determine which specific sub-skills the virtual fly has acquired during its interaction with the app, the system employs the Generalized Deterministic Input, Noisy "And" gate (G-DINA) model⁴⁷. The G-DINA model evaluates the agent's mastery profile over a set of discrete, latent cognitive attributes by analyzing its binary success or failure across various tasks⁴⁷.

In this experimental setup, the complex task of "reading" is deconstructed into fundamental biological and cognitive attributes via a $J \times K$ Q-matrix⁵⁰. For J reading tasks and K latent attributes, the matrix element $q_{jk} = 1$ if task j explicitly requires attribute k for successful completion, and 0 otherwise⁵¹. Table 2 illustrates a theoretical Q-matrix for the virtual fly, mapping complex behaviors to binary diagnostic attributes.

Item (Task) j	α1: Contrast Edge Detection	α2: Geometric Shape Discriminatio	α3: Sequential Symbol Tracking	α4: Precision Motor Targeting
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Task 1: Touch any dark region on screen	1	0	0	1
Task 2: Distinguish 'T' from 'H'	1	1	0	1
Task 3: Select 'A' followed by 'B'	1	1	1	1

The baseline DINA model is strictly conjunctive; it assumes that the fly will fail the task unless it possesses *all* the required attributes, subject only to a task-specific guessing probability (g_j) and a slipping probability (s_j)⁴⁷. The probability of success under the standard DINA model is defined as:

$$P(X_{ij} = 1 | \eta_{ij}) = (1 - s_j)^{\eta_{ij}} g_j^{1-\eta_{ij}}$$

where $\eta_{ij} = \prod_{k=1}^K \alpha_{ik}^{q_{jk}}$ represents whether the individual has mastered all necessary skills for the item⁵⁵.

However, because the highly interconnected neural dynamics of the FlyGM connectome may allow for partial mastery or compensatory neural pathways, the strictly conjunctive assumption of the standard DINA model is often too restrictive. The saturated G-DINA model provides a more flexible parameterization⁵¹. The G-DINA partitions the 2^K possible latent classes into $2^{K_j^*}$ distinct groups, where K_j^* is the number of attributes specifically required by item j ⁴⁸. Using an identity link function, the Item Response Function (IRF) for the G-DINA model is expressed as:

$$P(X_j = 1 | \alpha_{ij}^*) = \delta_{j0} + \sum_{k=1}^{K_j^*} \delta_{jk} \alpha_{ik} + \sum_{k'=k+1}^{K_j^*} \sum_{k=1}^{K_j^*-1} \delta_{jkk'} \alpha_{ik} \alpha_{ik'} + \cdots + \delta_{j12 \dots K_j^*} \prod_{k=1}^{K_j^*} \alpha_{ik}$$

where δ_{j0} is the baseline probability of success (analogous to guessing), δ_{jk} represents the main effect of possessing attribute k , and the subsequent terms denote higher-order interaction effects between the attributes⁴⁸. Logit and log link functions can similarly be applied to constrain the saturated model into specialized, parsimonious forms such as the Linear Logistic Model (LLM) or the Reduced Reparameterized Unified Model (R-RUM)⁴⁹.

Parameter Estimation, Identifiability, and Item Fit

Model parameters are calibrated using Marginal Maximum Likelihood (MML) implemented via the Expectation-Maximization (EM) algorithm running continuously on the LRS servers⁴⁸. In the E-step, the posterior probability of each latent attribute profile is computed given the fly's

kinematic response sequence; in the M-step, the structural parameters (δ) and the population prevalences are updated to maximize the likelihood⁴⁷.

A critical concern in cognitive diagnosis is parameter identifiability—ensuring that the model parameters can be uniquely recovered from the observed data⁶⁴. Strict identifiability requires highly specific Q-matrix designs that may not naturally occur in a continuous physics simulation⁶⁴. However, recent advances in psychometrics demonstrate that the DINA framework benefits from generic identifiability, meaning parameters are identifiable almost everywhere in the parameter space, excluding only negligible subsets of measure zero⁶⁴. More importantly, the evaluation leverages a phenomenon known as the "blessing-of-latent-dependence"⁶⁴. Because

the physical execution of navigation (α_4) and the visual processing of shape (α_2) are highly statistically dependent within the interconnected FlyGM network, this latent dependence actually restores mathematical identifiability in otherwise unidentifiable, sparse Q-matrix settings⁶⁴.

To ensure the hypothesized Q-matrix accurately reflects the cognitive demands placed on the

FlyGM network, model fit is evaluated using the $S - X^2$ item fit statistic⁵⁶. The $S - X^2$ compares the observed and expected response proportions across score groups, calculated as:

$$S - X_j^2 = \sum_{k=1}^{J-1} N_k \frac{(O_{jk} - E_{jk})^2}{E_{jk}(1 - E_{jk})}$$

If an interaction task exhibits a high $S - X^2$ value, indicating significant misfit, it suggests that the assigned attributes in the Q-matrix do not align with the actual sensorimotor processing mechanisms being utilized by the fly's simulated nervous system, prompting an automated adjustment of the curriculum⁵⁶.

Longitudinal Memory and Spaced Repetition

While Cognitive Diagnosis Models evaluate static mastery at a given moment, learning is fundamentally a temporal process. To map the longitudinal acquisition and retention of reading skills, the system integrates Dynamic Bayesian Networks (DBNs) alongside Half-Life

Regression.

Knowledge Tracing via Half-Life Regression (HLR)

To schedule the optimal presentation of language tasks and quantify memory retention, the system employs Half-Life Regression (HLR). HLR is a foundational algorithm used in the actual Duolingo platform's learner modeling, designed to operationalize spaced repetition⁶⁹. HLR marries the psycholinguistic theory of exponential memory decay—popularized as the Ebbinghaus forgetting curve—with supervised machine learning⁶⁹.

HLR models the probability p of the fly correctly recalling a visual association (e.g., distinguishing a target symbol from a newly introduced distractor) based on the time elapsed since the last exposure to that symbol (Δ) and the inferred strength of the memory in the fly's neural weights, termed the half-life (h)⁷⁰. The relationship is mathematically defined by the decay function:

$$p(\Delta, h) = 2^{-\Delta/h}$$

Instead of assuming fixed intervals for task scheduling, HLR treats the half-life h as a learnable quantity parameterized by a feature vector x . This vector summarizes the fly's review history, capturing variables such as the number of correct interactions, the number of incorrect interactions, and the inherent visual complexity of the symbol⁷¹. The estimated half-life $\hat{h}$ is calculated using an exponential link function:

$$\hat{h}(x) = 2^{\Theta \cdot x}$$

where Θ is a vector of learned weights mapped to the features⁷¹. For example, if the dot product increases by a factor of 1 due to a successful trial (which triggers a dopamine reward in the simulated PAM cluster), the half-life effectively doubles, stretching the forgetting timescale⁷¹. Conversely, if the fly fails the interaction, the half-life is compressed. By continuously running HLR inference on the xAPI telemetry stream, the system optimizes the presentation of new symbols just as the probability of recall for a previous symbol approaches the 50% threshold, ensuring maximum learning efficiency⁷¹.

State Tracking with Dynamic Bayesian Networks (DBN)

To complement the macro-level memory tracking of HLR, Dynamic Bayesian Networks (DBNs) are employed for granular, multi-skill knowledge tracing⁷⁴. A DBN extends classical Bayesian networks to model sequential data over time, capturing the complex conditional dependencies between different variables⁷⁵.

In this framework, a DBN is defined by a Directed Acyclic Graph (DAG) that models the joint distribution of hidden knowledge states (the fly's true mastery of specific graphemes) and

observed evidence (kinematic efficiency, path tortuosity, and correct selections) across consecutive time slices⁷⁴. The DBN defines transition probabilities representing the likelihood

that the fly transitions from a state of non-mastery to mastery between trial t and $t + 1$, and emission probabilities representing the likelihood of an observed physical action given the current hidden state⁷⁴.

For continuous kinematic observations derived from the MuJoCo simulation, the emission probability can be modeled using a mixture of Gaussian distributions, where the mean μ and covariance Σ change depending on the underlying cognitive state⁷⁷. Unlike standard Hidden Markov Models (HMMs), which rely on a single discrete latent variable, DBNs factorize the hidden state⁷⁴. This allows the model to independently trace the parallel acquisition of distinct attributes—such as geometric shape recognition and precision motor targeting—simultaneously, providing a holistic view of the agent's cognitive development⁷⁴.

Evaluating Simulation and Task Variance: The Many-Facet Rasch Model

As the simulated fly generates longitudinal data, a critical analytical challenge emerges: disentangling the inherent difficulty of the generated language tasks from the intrinsic learning capability of the connectome. Furthermore, "success" in a continuous physics simulation is rarely binary. It is judged by categorical rubrics evaluating motor efficiency, response latency, and targeting accuracy, introducing multiple sources of measurement variance⁷⁹.

The Many-Facet Rasch Model (MFRM), an extension of standard Item Response Theory (IRT) developed by Linacre, solves this problem by mathematically projecting multiple facets of the testing situation onto a single, objective, linear logit scale⁸⁰.

In this application, the facets analyzed by the MFRM include:

1. **Person Ability** (θ_p): The underlying learning capacity of the specific neural policy being tested (e.g., comparing the biological FlyGM against a purely synthetic xLSTM baseline).
2. **Item Difficulty** (b_i): The inherent visual and navigational difficulty of the specific reading task presented on the app.
3. **Rater Severity** (c_r): The strictness of the evaluation metric (e.g., an automated "strict" metric requiring millimeter-precision targeting on the screen versus a "lenient" metric allowing general proximity selection).

The MFRM defines the log-odds of a successful interaction as a linear, additive function of these independent parameters⁸¹. For a rating scale application where performance is graded across multiple categories, the model is formulated as:

$$\ln \left(\frac{P_{pirk}}{P_{pir(k-1)}} \right) = \theta_p - b_i - c_r - d_k$$

where P_{pik} is the probability of agent p receiving a performance score of k on item i under metric r , and d_k represents the step difficulty of transitioning from score category $k - 1$ to k ⁸³.

By treating the MFRM mathematically as a generalized linear mixed model (GLMM), parameter estimation is executed via crossed random effects, formulated as

$\text{logit}(P) = \beta_0 + u_p + v_i + w_r$ ⁸¹. The critical advantage of the MFRM is its ability to construct truly "fair" measures⁸⁰. If a particular language task appears artificially difficult simply

because the collision detection threshold within the MuJoCo engine (the rater severity, c_r) was set too strictly, the MFRM mathematically isolates and controls for this severity variation⁸¹.

This normalization enables researchers to generate a Wright Map. A Wright Map is a unified visual scale aligning the virtual fly's true cognitive ability directly against the calibrated difficulty of the Duolingo tasks and the strictness of the evaluation metrics⁸⁰. This provides an absolute, objective measurement of learning that is independent of the local quirks and tuning details of the physics engine.

Conclusion

The proposition to evaluate an open-source, simulated *Drosophila melanogaster* as it physically interacts with a web-based language acquisition platform represents a profound paradigm shift in Embodied Artificial Intelligence and cognitive psychometrics. By integrating the anatomically precise MuJoCo flybody with the biologically faithful FlyGM connectome, researchers are provided with an unprecedented, high-fidelity testbed for exploring sensorimotor integration and the biological roots of learning.

To systematically and rigorously evaluate the fly's ability to engage in proto-reading, this framework establishes the absolute necessity of a multi-tiered, highly integrated algorithmic approach. The infrastructure relies on browser-based WebAssembly execution coupled with xAPI telemetry streaming over a Science DMZ, ensuring the uninterrupted capture of high-frequency behavioral and neural data. Sequence modeling, driven by the matrix memory of the xLSTM and the offline reinforcement logic of Decoupled Decision Transformers, provides the high-capacity temporal memory and adaptive curriculum sequencing required to navigate language tasks.

Simultaneously, the psychometric evaluation is anchored by Half-Life Regression, which precisely maps the exponential decay of learned symbols, while the G-DINA model, via MML-EM parameter estimation, provides granular profiles of the agent's latent cognitive attributes, relying on latent dependence to ensure parameter identifiability. Finally, the Many-Facet Rasch Model standardizes the varying complexities of the digital tasks, the automated evaluation metrics, and the neural policies onto a singular, objective logit scale. Ultimately, this synthesis of computational neuroscience, large-scale deep learning, and advanced psychometrics yields an evaluative framework that moves far beyond rudimentary reinforcement learning metrics. It provides a robust, scientifically valid mechanism for

quantifying the emergence of complex cognitive behaviors in biologically constrained artificial systems, answering the fundamental questions posed by the Digital Sphinx.

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